Algoritmo_Genetico_para_el_TSP

October 15, 2026

#Instalación de librerias

[ ]:
```
!pip install requests
import urllib.request

!pip install tsplib95
import tsplib95
```

[ ]:
```
import random   #Libreria para generar numeros y listas aleatorias
import copy     #Permite hacer copias de objetos en python: listas,␣
 ↪diccionarios,...
```

#Carga de datos del problema

[ ]:
```
#Librerias y carga del problema

#http://elib.zib.de/pub/mp-testdata/tsp/tsplib/
#Documentacion :
  # https://wwwproxy.iwr.uni-heidelberg.de/groups/comopt/software/TSPLIB95/
 ↪tsp95.pdf
  # https://tsplib95.readthedocs.io/usage.html
  # https://tsplib95.readthedocs.io/modules.html#module-tsplib95.models

#Matriz de distancias
file = "swiss42.tsp" ;
urllib.request.urlretrieve("http://comopt.ifi.uni-heidelberg.de/software/
 ↪TSPLIB95/tsp/swiss42.tsp.gz", file + '.gz')
!gzip -d swiss42.tsp.gz     #Descomprimir el fichero de datos

#Objeto de tsplib95 para nuestro problema problema
problem = tsplib95.load(file)

#Nodos
Nodos = list(problem.get_nodes())

#Aristas
Aristas = list(problem.get_edges())
print(Aristas)
```

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#Coordenadas(si estan disponibles en el ficher)
problem.get_display(1)

#Distancia
problem.get_weight(1, 2)
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[ ]:
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#Funciones de la Actividad Guiada 3

[ ]:
```
#Funciones de la Actividad Guiada 3
#Se genera una solucion aleatoria con comienzo en en el nodo 0
def crear_solucion(Nodos):
  solucion = [Nodos[0]]
  for n in Nodos[1:] :
    solucion = solucion + [random.choice(list(set(Nodos) - set(solucion)))]
  return solucion

#Devuelve la distancia entre dos nodos
def distancia(a,b, problem):
  return problem.get_weight(a,b)


#Devuelve la distancia total de una trayectoria/solucion
def distancia_total(solucion, problem):
  distancia_total = 0
  for i in range(len(solucion)-1):
```

```
    distancia_total += distancia(solucion[i] ,solucion[i+1] ,  problem)
  return distancia_total + distancia(solucion[len(solucion)-1] ,solucion[0],␣
 ↪problem)
```

[ ]:

#Funciones Auxiliares

[ ]:
```
#Genera una poblacion inicial de soluciones de tamaño N.
# Puede ser válida la solución aleatoria de la AG3: crear_solucion(Nodos)
def generar_poblacion(Nodos,N):
  #
```

[ ]:
```
#Evalua la población y devuelve el mejor individuo
def Evaluar_Poblacion(poblacion, problem):
  #...
```

[ ]:
```
#Funcion de cruce. Recibe una poblacion(lista de soluciones) y devuelve la␣
 ↪población ampliada con los hijos.
# Todos los individuos de la población son selecionados para el cruce(si la␣
 ↪población es par)
# Podría aplicarse un proceso previo de selección para elegir los individuos␣
 ↪que se desea cruzar.
def Cruzar(poblacion,mutacion, problem):
  #...
```

[ ]:
```
#Funcion para generar hijos a partir de 2 padres:
# Se elige el metodo de 1-punto de corte pero es posible usar otros n-puntos,␣
 ↪uniforme, dependiendo del problema
def Descendencia(padres, problem,mutacion):
  #...
```

[ ]:
```
#Para el operador de cruce 1-punto los hijos generados no son␣
 ↪soluciones(algunos nodos se repiten y otros no están)
def Factibilizar(solucion,problem):
  #...
```

[ ]:
```
#Funcion de mutación. Se eligen dos nodos y se intercambia. Se podrian añadir␣
 ↪otros operaradores
# Se hace mutaciones mutacion% de las veces
def Mutar(solucion,mutacion):
  #
```

[ ]:
```
#Funcion de seleccion de la población. Recibe como parametro una poblacion y
# devuelve una poblacion a la que se ha eliminado individuos poco aptos(fitness␣
 ↪alto) y para mantener una poblacion estable de N individuos
#Se tiene en cuenta el porcentaje elitismo pasado como parametro
```

```
# Para los individuos que no son de la elite podríamos usar una selección de
 ↪ruleta(proporcional a su fitness)
def Seleccionar(problem,poblacion, N, elitismo):
```

[ ]:

#Proceso Principal

[ ]:
```
#Funcion principal del algoritmo genetico
############################################################3
def algoritmo_genetico(problem=problem,N=100,mutacion=.15,elitismo=.
 ↪1,generaciones=100):
  # problem = datos del problema
  # N = Tamaño de la población
  # mutacion = probabilidad de una mutación
  # elitismo = porcion de la mejor poblacion a mantener
  # generaciones = nº de generaciones a generar para finalizar

  #Genera la poblacion inicial
  Nodos = list(problem.get_nodes())
  poblacion = generar_poblacion(Nodos,N)

  #Inicializamos valores para la mejor solucion
  (mejor_solucion, mejor_distancia) = Evaluar_Poblacion(poblacion, problem)


  #Condicion de parada
  parar = False
  n=0
  #Inciamos el cliclo de generaciones
  while(parar == False) :

    #Cruce de la poblacion(incluye mutación)
    poblacion = Cruzar(poblacion, mutacion, problem)

    #Seleccionamos la población
    poblacion = Seleccionar(problem,poblacion, N, elitismo)

    #Evaluamos la nueva población
    (mejor_solucion, mejor_distancia) = Evaluar_Poblacion(poblacion, problem)

    print("Generacion #", n, "\nLa mejor solución es:" , mejor_solucion, "\ncon
 ↪distancia " , mejor_distancia, "\n")

    #Numero de generaciones. Criterio de parada
    if n==generaciones:
      parar = True
```

```
    n +=1

  return mejor_solucion


sol = algoritmo_genetico(problem=problem,N=500,mutacion=.3,elitismo=.
 ↪40,generaciones=250)
```